Database Management Systems Project

Team Members:

B231202018	Elif Yırtıcı	elif.yirtici@ogr.sakarya.edu.tr
B231202036	Harun İlker Kaya	harun.kaya4@ogr.sakarya.edu.tr
B231202067	Nurgül Dinçer	nurgul.dincer@ogr.sakarya.edu.tr
B251202350	Ayşe Dönmez	ayse.donmez3@ogr.sakarya.edu.tr

Travel Information Management System (TMS)

Scenario

• A travel management system is requested to design a database for organizing tourism activities.
• In the designed database; travelers, guides, trips, accommodations, touristic places, reviews, countries, and travel companies are expected to be stored.
• The system should manage travel histories, guide assignments, visited places, and traveler feedback in a consistent and relational structure.

Business Rules

Business Rules

• In the system, each person has a unique person Id, name, email, and phone number. Additionally, a person can exist in only two categories: a traveler or a tour guide.
A person cannot be both at the same time. traveler and a tour guide have also some specific attributes which are written in the next business rules.

• Traveler information should include passport number. Tour guide information should include experience year.

•Each trip has a unique trip Id, trip period .

• A guide can participate in many trips or no trip. A trip is organized by only one guide. There can't be any trip that doesn't related to a tour guide.
•A traveler take at least one trip but can also take many trips. A trip can be taken by one or many travelers.

• Reviews written by travelers must be stored with a unique review ID, rating, and comment.

• A traveler can write multiple reviews or no review. A review must be written by only one traveler.

• A trip can have many reviews or no reviews. A review must belong to only one trip.Reviews can not be existed without a related trip.

•Travel companies' ID numbers and company names must be stored.

•Each Tour Guide must be associated with exactly one Travel Company. A Tour Guide cannot exist in the system without a corresponding Travel Company record. A company must include at least one tour guide, but it may include many tour guides.

•Touristic places have unique place Ids, names and the related country information stored.

•Countries have unique country Ids and country names.

•A touristic place must belong to only one country. A country can have many touristic places but must have at least one. There cannot be any touristic place that does not belong to a country.

• A touristic place can appear in multiple trips or never be visited. A trip must include at least one touristic place, but it may include multiple touristic places. For every touristic place visited during a trip, the system must store the corresponding visit date

• A trip must belong to only one travel company. A travel company can manage many trips or no trip.There can't be any trip that doesn't related to a company.

• Accommodations' ID numbers, names, addresses and related countries must be stored.

•Every accommodation must belong to exactly one travel company. A travel company can have many accommodations or no accommodation. Accommodations cannot exist independently without being associated with a company.

• Each accommodation must belong to a country. A country can have many accommodations or none. An accommodation cannot exist without country.

•A trip may have one accommodation or none. An accommodation can be linked to many trips or no trip.

• A trip must belong to exactly one destination country . A country can have many trips or no trips at all. There can't be any trip that doesn't have destination country.

Relational Schema (Textual Representation) :

•Person(personId:char,name:varchar,email:char,phoneNumber:varchar,personType:char)

• Traveler(personId: char,passportNumber:char)

• TourGuide(personId: char, companyId: char,experienceYear:int)

• Company(companyId: char, companyName: varchar)

• Accommodation(accomodationId: char,name: varchar, address: varchar, country: varchar, companyId: char)

• Country (countryId:char , name:varchar)

• TouristicPlace (placeId:char, name:varchar, countryId:char)

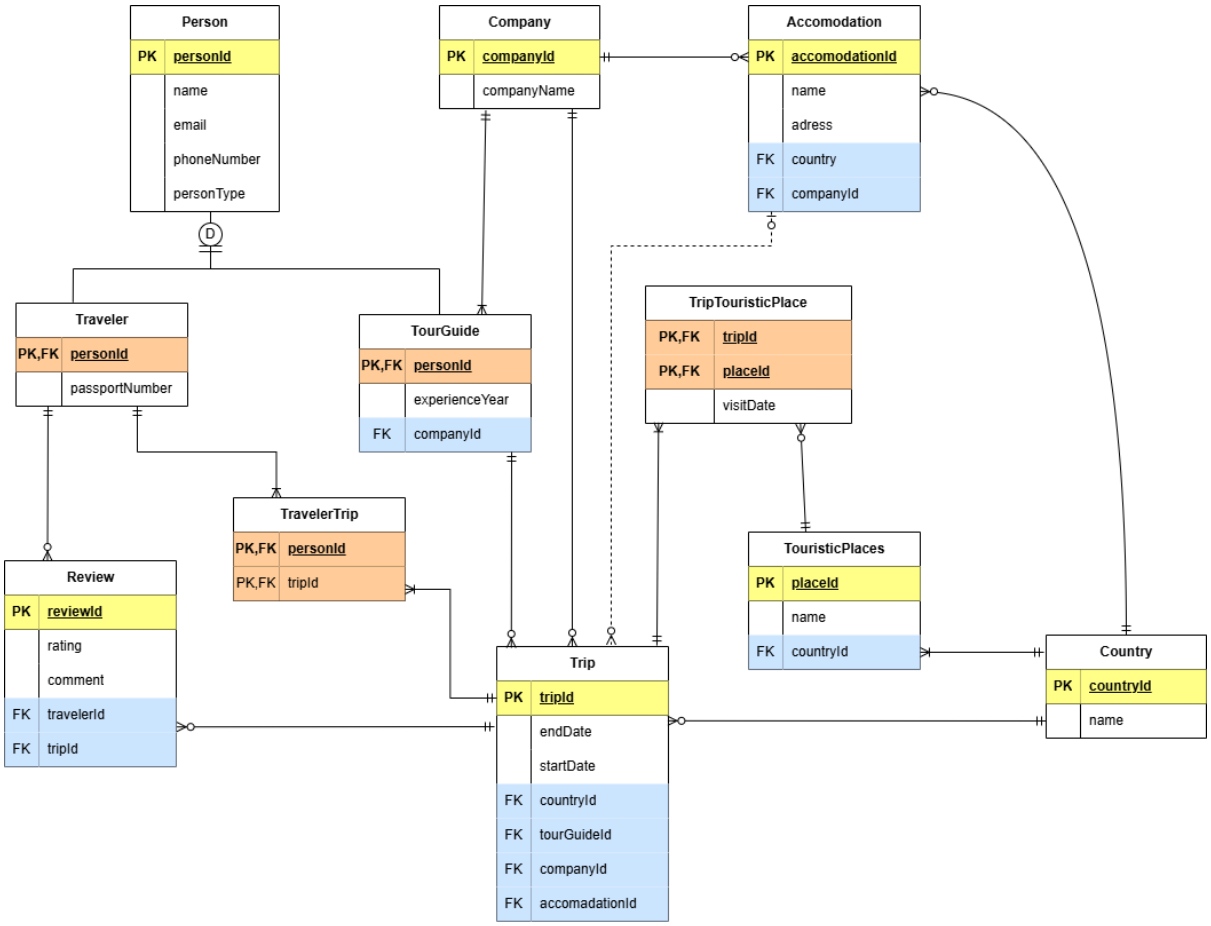
•Trip(tripId:char,startDate:date,endDate:date,countryId:char,tourGuideId:char,companyId:char,accommodationId:char)

• TripTouristicPlace (tripId:char, placeId:char , visitDate:date)

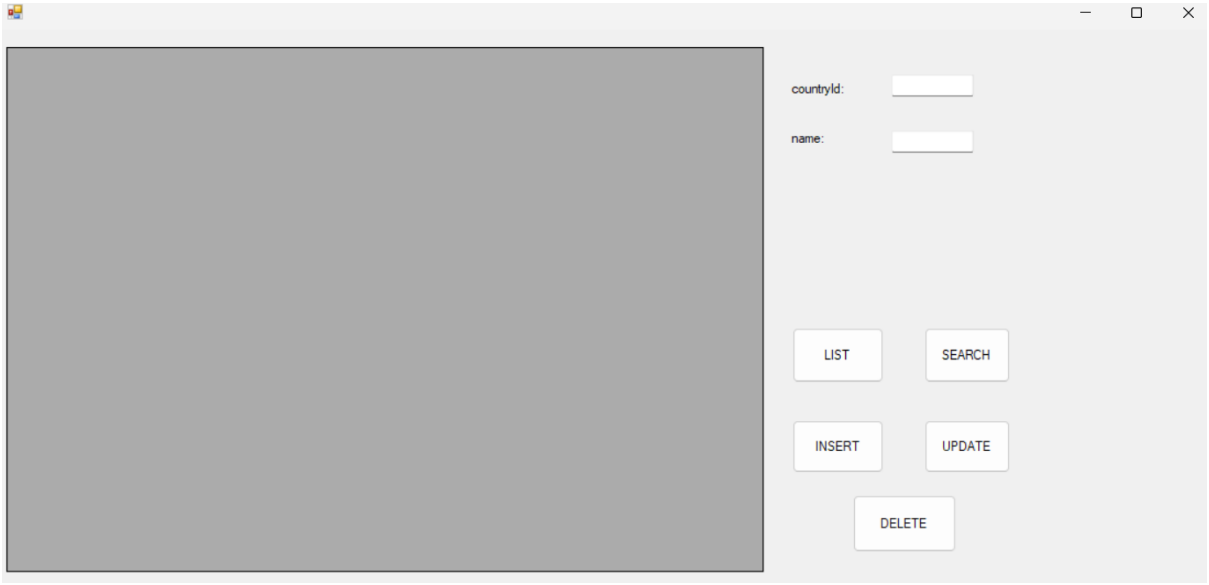
• Review (reviewId:char, rating:int, comment:varchar, travelerId:char , tripId:char)

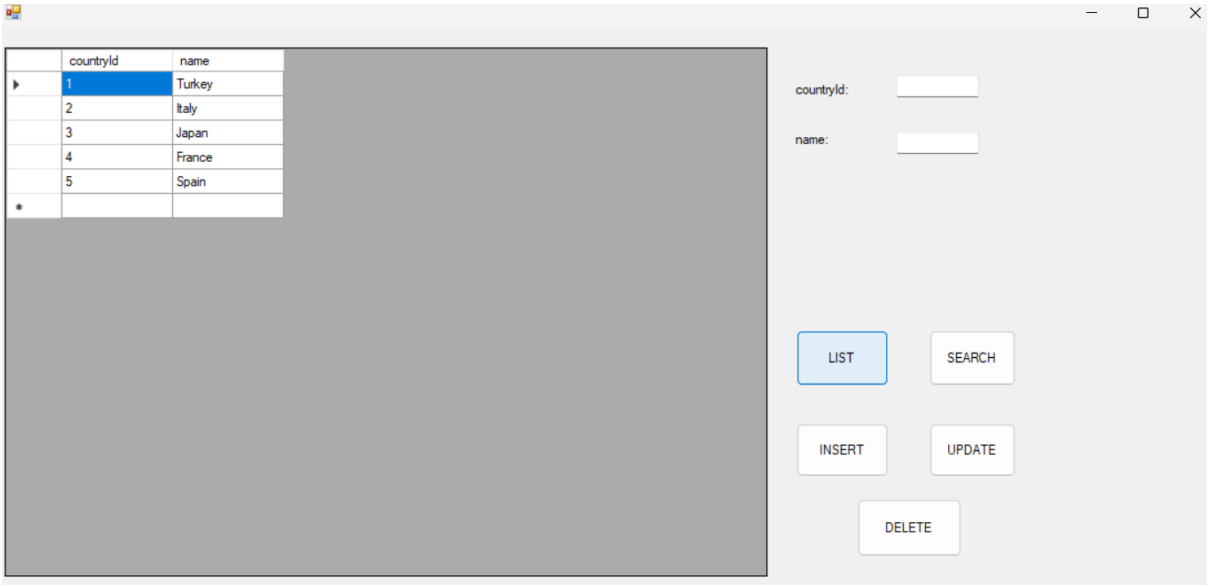
•TravelerTrip(personId:char,tripId:char)

Entity Relationship Model (Crow's Foot) Notation



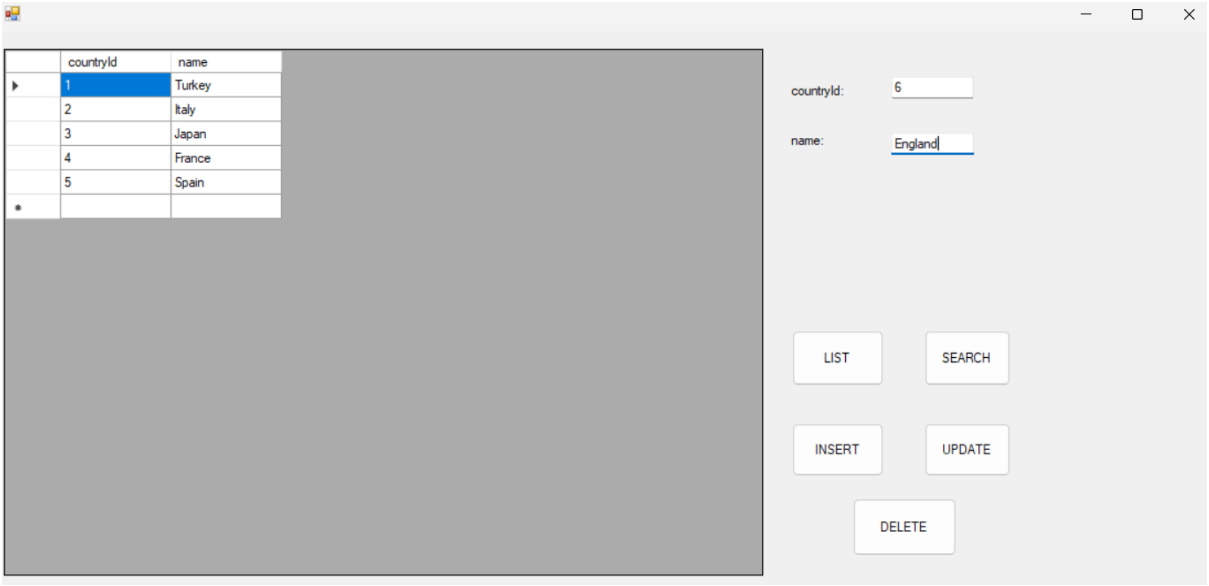
- **Screenshots of List**

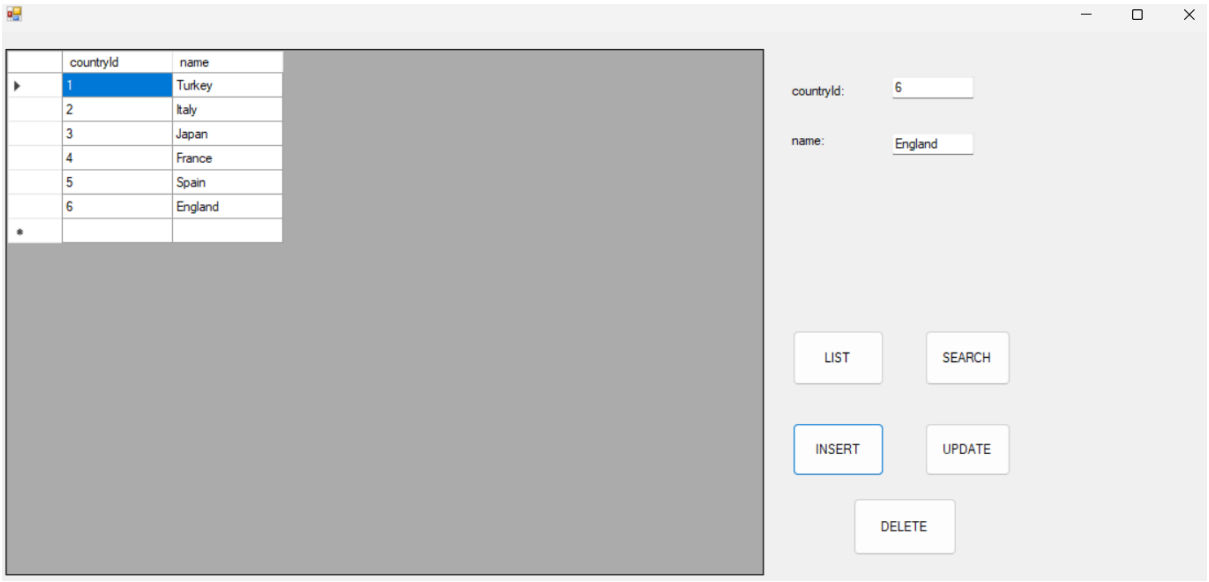




All registered country records are retrieved from the database and displayed in the data grid interface.

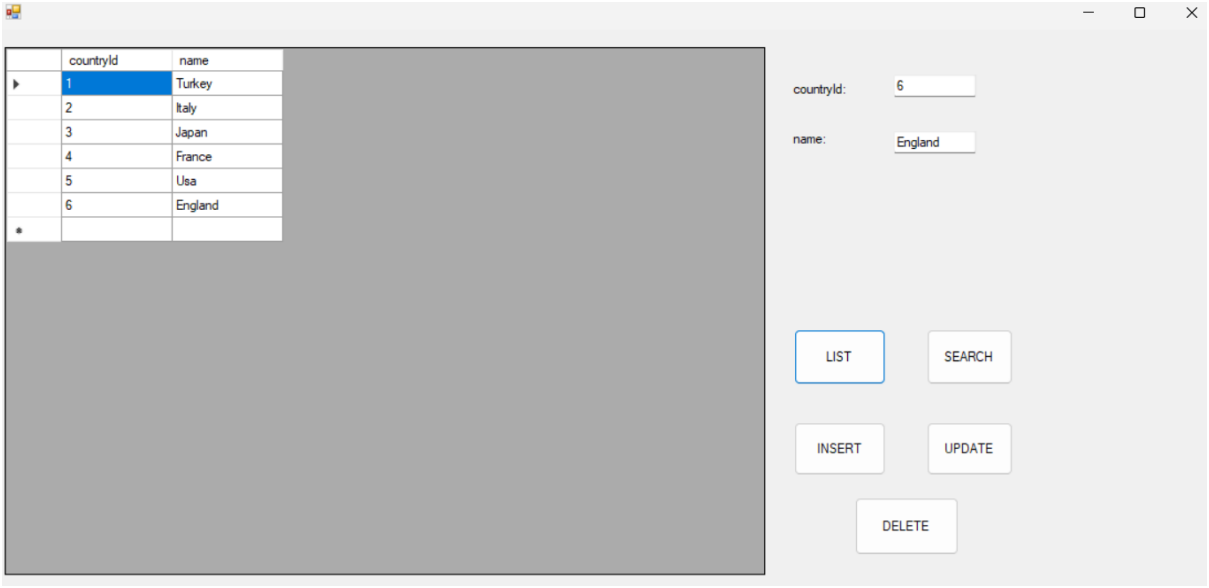
- **Screenshots of Insert**

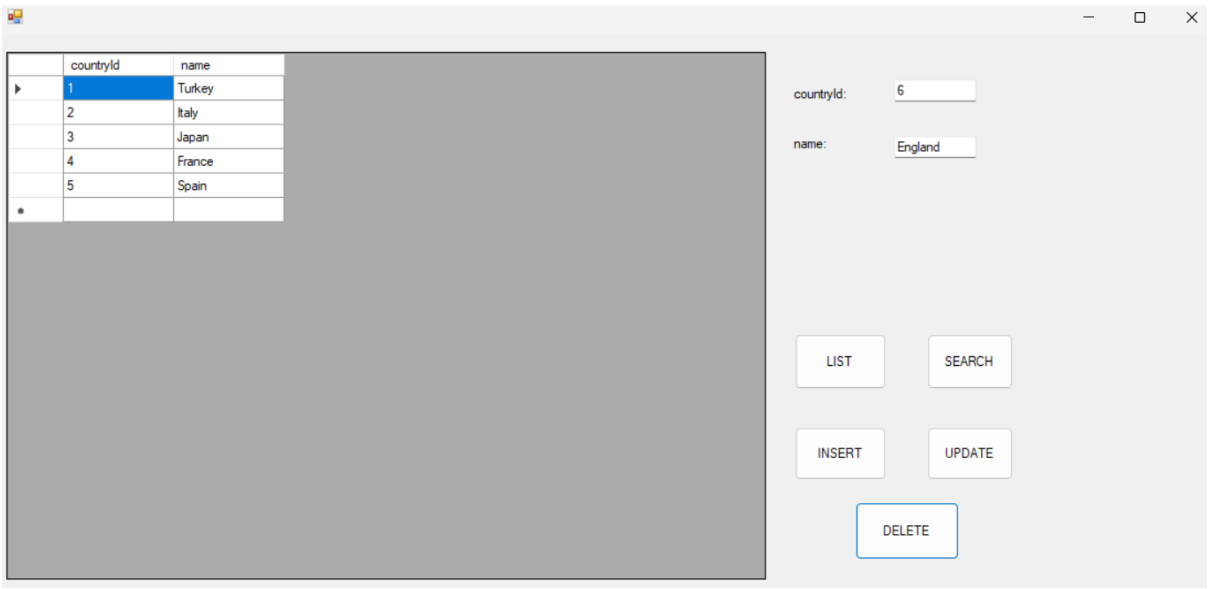




New country information is entered into the system fields and added to the database records.

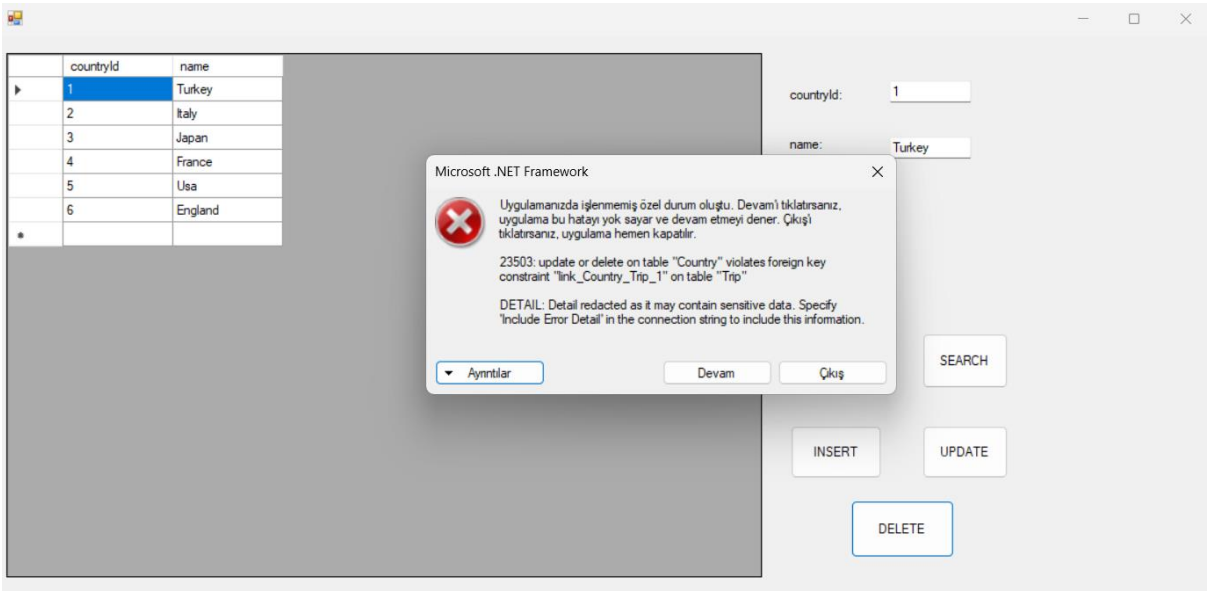
- **Screenshots of Delete**





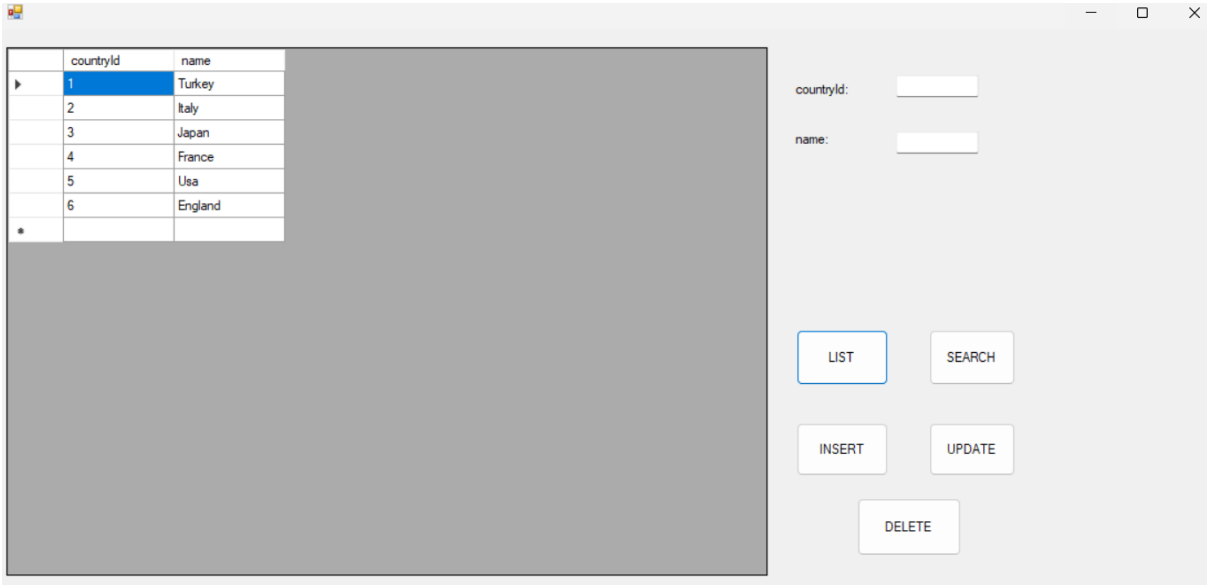
A specific country record is permanently removed from the system by targeting its unique countryId.

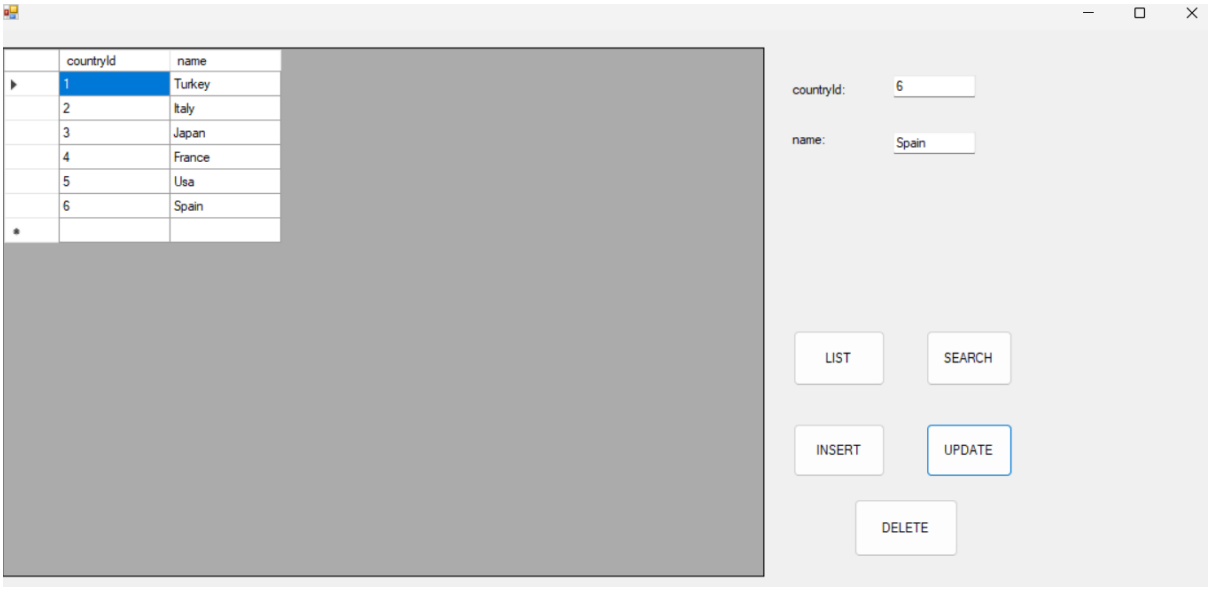
Data Integrity and Foreign Key Constraint Test



During the data deletion tests performed on the system, the behavior of the database was observed when attempting to delete a record that has active relationships with other tables.In the example shown in the screenshot, an attempt to permanently remove a record from the 'Country' table resulted in a PostgreSQL 23503 error. This error indicates that the specific country is currently referenced by active records in the 'Trip' table.

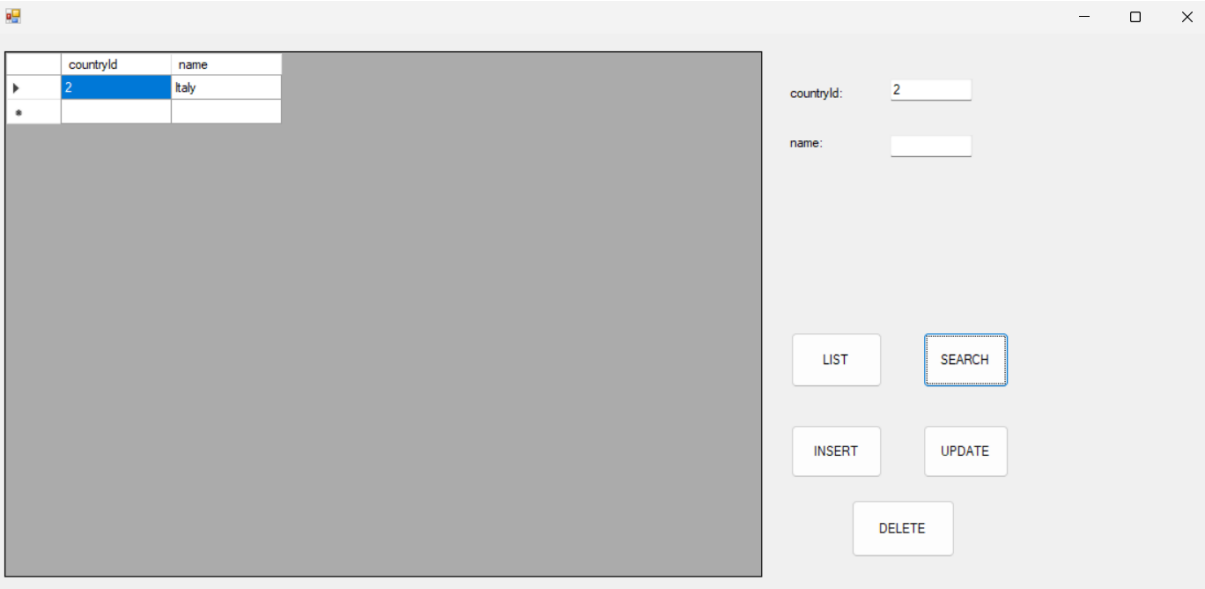
- **Screenshots of Update**





Existing country information is modified via the interface and the changes are saved back to the database.

- **Screenshots of Search**



A specific country record is retrieved from the database by querying its unique countryId.

DBMS_Project.sql : Our sql statements.